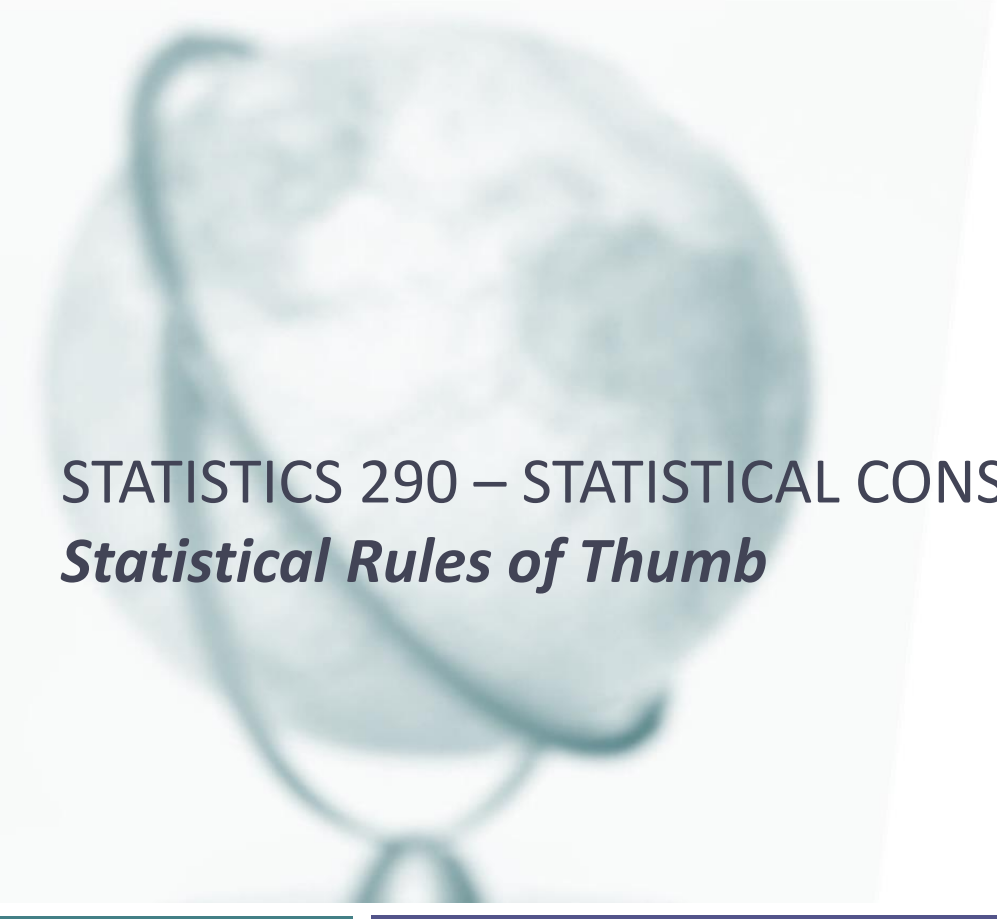

STATISTICS 290 – STATISTICAL CONSULTING

Statistical Rules of Thumb

Statistical Rules of thumb

- Van Belle, G., 2008, Statistical Rules of Thumb, New Jersey: Wiley
  - The Basics
  - Sample Size
  - Observational Studies
  - Covariation
  - Design, Conduct, and Analysis
  - Consulting

The Basics

- Rule of Thumb
  - Any statistical treatment must address the following questions:
    - What is the question?
    - Can it be measured?
    - Where, when, and how will you get the data?
    - What do you think the data are telling you?
- Basis of the Rule
  - Mimics the scientific process with particular emphasis on the role of data collection and analysis.

The Basics

- Rule of Thumb
  - Observation is selection.
- Basis of the Rule
  - To observed one thing implies that another is not observed, hence there is selection, implying that observation is taken from a larger collective, the statistical “population”.

The Basics

- Rule of Thumb
  - Replicate to characterize random variation
- Basis of the Rule
  - Characterizing variability requires repeatedly observing the variability since it is not a property inherent in the observation itself.

The Basics

- Rule of Thumb
  - Variability occurs at multiple levels.
- Basis of the Rule
  - Recognition that there are levels of variations.

The Basics

- Rule of Thumb
  - Invalid selection is the primary threat to valid inference.
- Basis of the Rule
  - Selection bias influences outcomes of statistical inference.

The Basics

- Rule of Thumb
  - Compared with experimental studies, observational studies provide less robust information.
- Basis of the Rule
  - Reason for less robust inference of observational studies: probability framework linking the study to the population of inference in unknown.

The Basics

- Rule of Thumb
  - Make a sharp distinction between observational and experimental studies.
- Basis of the Rule
  - A probability model is required to draw valid inferences.

The basics

- Rule of Thumb
  - Always look for a physical model underlying the data being analyzed. Assume that a statistical model such as a linear model is a good first start only.
- Basis of the Rule
  - It is crucial to have a broad understanding of the subject matter involved. Statistical analysis is much more than just carrying out routine comparisons. Only with keen understanding of the subject matter can statisticians, and statistics, be most usefully engaged.

The Basics

- Rule of Thumb
  - Keep models as simple as possible, but not more simple.
- Basis of the Rule
  - Simpler is better. Parsimony. Einstein: Explanations of scientific phenomena should be as simple as possible, but not more simple!

The Basics

- Rule of Thumb
  - Be sure to understand the components and purpose of an omnibus quantity.
- Basis of the Rule
  - Omnibus quantities represents many-one mappings. Thus, for the same value of this quantity many component values are possible. Ordinarily the components are of primary scientific interest, while the omnibus quantity is a shorthand summary of intermediate interest.

The Basics

- Rule of Thumb
  - The use of one-sided p-values is discouraged. Ordinarily, use two-sided p-values.
- Basis of the Rule
  - If the null hypothesis is false, it is not known a priori which direction the relationship will go.

The Basics

- Rule of Thumb
  - When designing experiments or observational studies focus on p-values to calculate sample size; **when reporting results focus on confidence intervals**.
- Basis of the Rule
  - In hypothesis testing, the investigator “stands” on the null hypothesis and see where the statistics is located. In estimation, the investigator is located on the statistic and see where the hypothesis values is located. In both cases, p-value is equivalent to the distance between the hypothesis and the statistic.

The Basics

- Rule of Thumb
  - The width of a confidence interval involving estimation of variability and sample size decreases rapidly until 12 observations are reached and then decreases less rapidly.
- Basis of the Rule

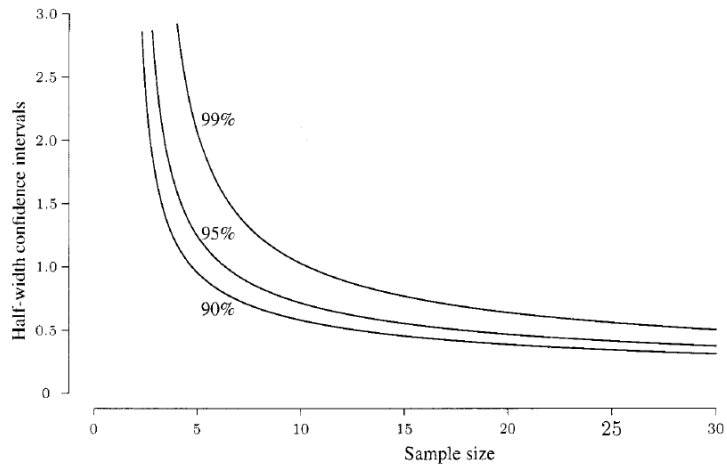

Fig. 1.1 Half–width confidence interval assuming a t–statistic with $n-1$ degrees of freedom and sample size n. Confidence levels of 90%, 95%, and 99%.

The Basics

- Rule of Thumb
  - Always know what the unit of a variable is.
- Basis of the Rule
  - Defining units explicitly is a scientific courtesy that fosters sound communication.

The Basics

- Rule of Thumb
  - Do not let the scale of measurement rigidly determine the method of analysis.
- Basis of the Rule
  - Most of the time variables are mixtures or sums of diverse scales of measurement.

Sample Size

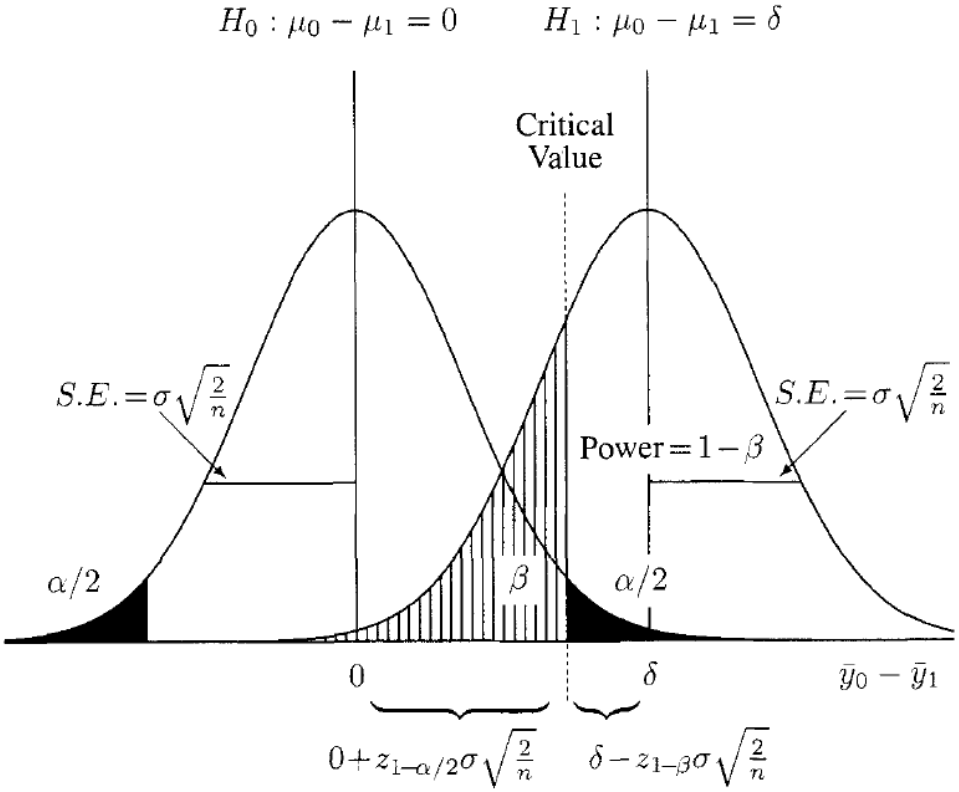

Fig. 2.1 Sampling model for two independent sample case. Two-sided alternative, equal variances under null and alternative hypotheses.

Sample Size

- Method of determination vary according to objective
  - In experiments, usually smaller size
    - Function of effect size and power
  - To estimate means, proportions/prevalence rate, usually bigger size
    - Function of margin of error, heterogeneity of population, sample selection
  - In modeling, lesser effect than configuration of x's

Experiments

- Control all factors across groups except the treatment.
- Process
  - Screen experimental units
  - Randomize
  - Apply the treatment
  - Measure response
  - Compare means
  - Differences in means (treatment effect) are used as evidence on effectiveness of treatment.

Experiments

- Small sample sizes: may be difficult to detect small effects
- Large sample sizes: may lead to contamination in dealing with large amount of data.
- Rationalize the size of the sample (replicates) to avoid the problems mentioned above.

Goal of Experiments

- To test: H_o:μ=μ_o vs. H_1:μ=μ_o
- H_o is the default, status quo
- Experiment:
  - Evidence to reject H_o
  - Or proof in favor of H_1
- Decision=correct or error
- Power: Chance of rejecting H_o when it is indeed false.

OC Curve

- The OC curves= effect of sample size to the power of the test.
  - Used in deciding a reasonable sample size.
- Note that a software like STATA is capable of plotting the OC curves
  - need not use the visually stressful graphs printed in many experimental design books

Example

. **sampsi 132.86 127.44, p(0.8) r(2) sd1(15.34) sd2(18.23)**

Estimated sample size for two-sample comparison of means

Test Ho: m1 = m2, where m1 is the mean in population 1

and m2 is the mean in population 2

Assumptions:

alpha = 0.0500 (two-sided)

power = 0.8000

m1 = 132.86

m2 = 127.44

sd1 = 15.34

sd2 = 18.23

n2/n1 = 2.00

Estimated required sample sizes:

n1 = 108

n2 = 216

```
. sampsi 132.86 97.44, p(0.8) r(2) sd1(15.34) sd2(18.23)

Estimated sample size for two-sample comparison of means

Test Ho: m1 = m2, where m1 is the mean in population 1
                      and m2 is the mean in population 2
Assumptions:

         alpha =    0.0500  (two-sided)
         power =    0.8000
            m1 =    132.86
            m2 =     97.44
           sd1 =     15.34
           sd2 =     18.23
         n2/n1 =      2.00

Estimated required sample sizes:

            n1 =         3
            n2 =         6
```

Example

. **sampsi 132.86 127.44, p(0.8) r(2) sd1(15.34) sd2(18.23)**

Estimated sample size for two-sample comparison of means

Test Ho: m1 = m2, where m1 is the mean in population 1

and m2 is the mean in population 2

Assumptions:

alpha = 0.0500 (two-sided)

power = 0.8000

m1 = 132.86

m2 = 127.44

sd1 = 15.34

sd2 = 18.23

n2/n1 = 2.00

Estimated required sample sizes:

n1 = 108

n2 = 216

```
. sampsi 132.86 127.44, p(0.8) r(2) sd1(8.34) sd2(9.23)

Estimated sample size for two-sample comparison of means

Test Ho: m1 = m2, where m1 is the mean in population 1
                    and m2 is the mean in population 2
Assumptions:

         alpha =    0.0500  (two-sided)
         power =    0.8000
            m1 =    132.86
            m2 =    127.44
           sd1 =      8.34
           sd2 =      9.23
          n2/n1 =     2.00

Estimated required sample sizes:

            n1 =        30
            n2 =        60
```

Example

```
. sampsi 5.6 6.1, n1(100) sd1(1.5) a(0.01) onesided
Estimated power for two-sample comparison of means
Test Ho: m1 = m2, where m1 is the mean in population 1
                and m2 is the mean in population 2
Assumptions:
       alpha =   0.0100  (one-sided)
          m1 =      5.6
          m2 =      6.1
         sd1 =      1.5
         sd2 =      1.5
sample size n1 =      100
            n2 =      100
       n2/n1 =     1.00
Estimated power:
       power =   0.5122
```

Estimation of Means/Proportions

- **What do you Need to Know?**
  - Goal of the Survey
  - Domain
  - Characteristics/variables of interest
  - Feasibility of Operations/Constraints/Timeline
  - BUDGET

Estimation of Means/Proportions

- **Sample Size in Element Sampling**
  - ***Sample size*** is the number of sampling units that will be drawn to represent the population.
  - It is determined during the planning stage of a survey.
  - Sample size determination uses information from a pilot study/pretest of previous studies/surveys.

Estimation of Means/Proportions

- **General Considerations**
  - as the sample size increases, cost also increases
  - if the population is heterogeneous, large sample is needed to represent it
  - if higher precision is desired, larger sample size should be taken
  - Large population, easy to represent, lower sample size
  - Small population, difficult to represent, larger sample size
  - Population Size affect Sample Size only if it is SMALL

Estimation of Means

- Given MOE(r)
- Example: C=SD/Mean=200/2000=0.1

  r=0.05

  $n=(0.1)^2 x (1.96)^2/(0.05)^2=15.4\approx16$

  To estimate calorie intake, it is known that the average is 2000 with SD of 200. With 95% confidence that the estimate will be within 5% of true intake, need 15 respondents.

Estimation of Proportions

Given r:

- Example:

  r=0.05, p=proportion with Vit. A deficiency=0.10

  $n=(0.1)(0.9)(1.96)^2/(0.05)^2=138.3=139$

  r=0.05, no idea of level, assume p=0.5

  $n=(0.5)(0.5)(1.96)^2/(0.05)^2=384$

r =	.10	.05	.04	.03	.02	.01
n =	96	384	600	1,067	2,401	9,604

Estimation of Parameters

- Defined only for SRS
- Effect of Stratification=Lower Variance
- Effect of multistage selection (clustering)=Higher Variance
  - Design effect is approximately 2
- MOE is target error
- Actual error can be lower/higher depending on actual selection, how much is known about the population.

Modeling

- n affects R^2 but not variance
- n depends on orthogonality of x's
- more observations near curvature points. Why?
- more observations far away from the mean. Why?

Purposive sampling

- Sample Size does not matter.
- Optimal design in Bayesian context.

Observational studies

- Things to consider according to Cochran
  - Setting up the comparisons (stratification vs. purposive selection)
  - Handling disturbing variables (confounders)-(now, observations as well)
  - Going from association to causation (more general structure)
  - Going from sample to population (compilation data)
  - Issues around measurement (aggregation, transformation)
  - Multiple variables and comparisons (high dimensionality)
  - Long-term studies (panel data, space-time interaction)

Observational studies

- Rule of Thumb
  - Model for an observational study is the sample survey
- Basis of the Rule
  - Surveys carefully specify the selection probabilities, observed data represents the population. Good surveys address specific questions formulated in the measurable way.

Observational studies

- Rule of Thumb
  - Large sample size does not guarantee validity.
- Basis of the Rule
  - Sample size determines precision not accuracy. Selection process determines accuracy (validity).

Observational studies

- Rule of Thumb
  - Good observational studies are designed.
- Basis of the Rule
  - Observational studies start farther back in the causal chain and therefore need extra care in planning and in analysis.

Observational studies

- Rule of Thumb
  - To establish cause and effect requires longitudinal data.
- Basis of the Rule
  - Since causes precede effects it requires observations measured over time.

Observational studies

- Rule of Thumb
  - “Make your theories elaborate”, i.e., consider many alternative explanations for the observed effect.
- Basis of the Rule
  - Any statistical relationship between two variables may be altered by additional variables.

Observational studies

- Rule of Thumb
  - Guidelines to ensure causation rather than association:
    - Strength (correlation. Test of association)
    - Consistency (association occurs in different situations)
    - Specificity (cause has specific effects)
    - Temporality (cause precede the effect)
    - Biologic gradient (dose-response curve)
    - Plausibility (theory, e.g., weight loss vs. l-carnitine)
    - Coherence (sensible, plausible)
    - Experimental Evidence (proof of concept)
    - Analogy (different situations that appear to produce similar effects)

Covariation

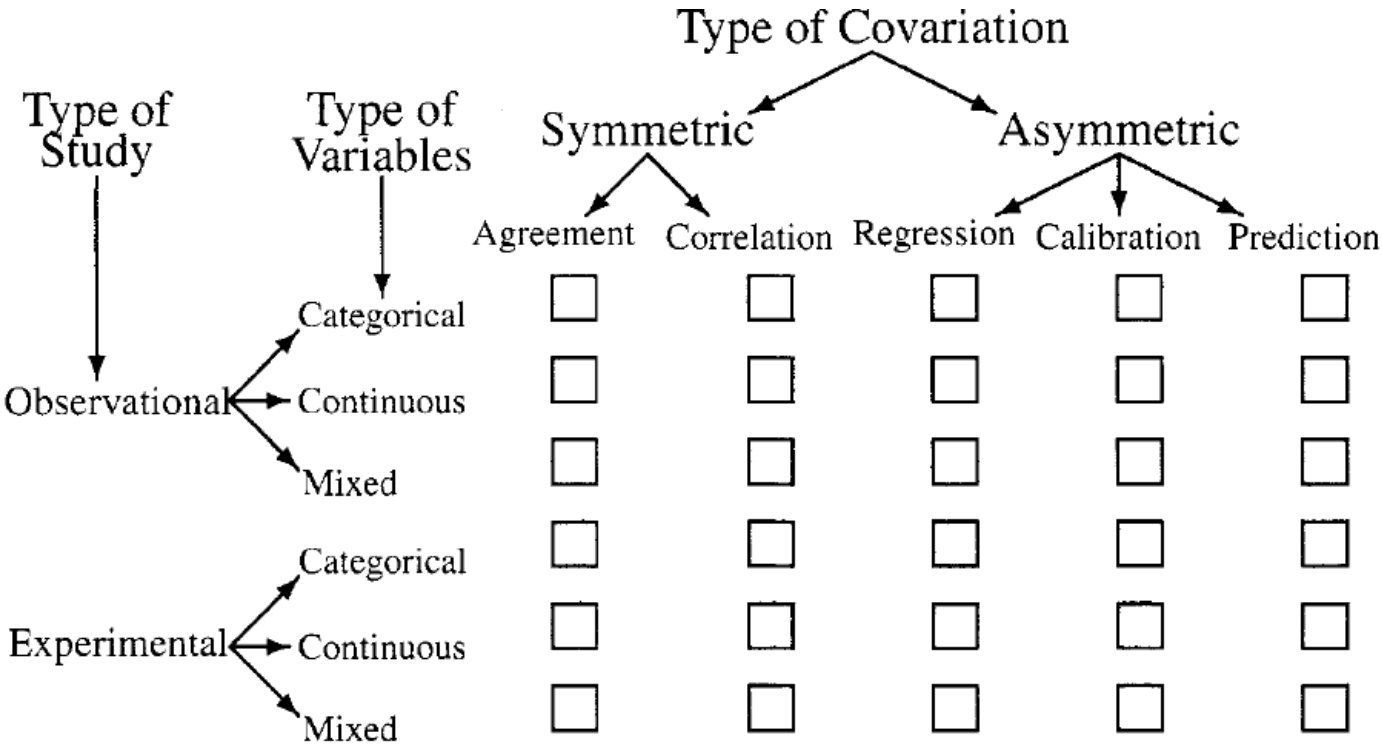

Fig. 4.1 Taxonomy of measures of covariation.

Covariation

- Rule of Thumb
  - Before choosing a measure of covariation determine the sources of data (sampling scheme), the nature of the variables, and the symmetry status of the measure.
- Basis of the Rule
  - Appropriateness of use: the wrong tools imply inappropriate assumptiions, and lead to incorrect or incomplete conclusions.

Covariation

- Rule of Thumb
  - Do not correlate rates of ratios indiscriminately.
- Basis of the Rule
  - Ratio variables with at least one common term are correlated!

Covariation

- Rule of Thumb
  - Pairing should be investigated carefully. Do not pair unless the correlation between the pairs exceed 0.5.
- Basis of the Rule

  Variance of differences of paired data

$$V(y_{i1} - y_{i2}) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

  Assuming equal variances

$$V(y_{i1} - y_{i2}) = 2\sigma(1 - \rho)$$

  If $\rho = 0$, variance of difference=sum of variances

  Variance of difference is reduced only if ρ is large.

Covariation

- Rule of Thumb
  - Do not use the coefficient of correlation as a measure of agreement when location and scale are relevant.
- Basis of the Rule
  - Invariance of the correlation coefficient under changes in location and scale is well known. Hence, effects attributable to location and differential variability will not be detected.

Design, conduct, and analysis

- Rule of Thumb
  - Randomization puts systematic sources of variability into the error term.
- Basis of the Rule
  - Purpose of randomization
    - Systematic effects are turned into error
    - Expected balance in the assignment of known and unknown factors that might influence the outcome.
    - Provides the basis of statistical procedures (sampling distribution).

Design, conduct, and analysis

- Rule of Thumb
  - By randomly assigning treatment to experimental units within blocks, systematic effects between blocks can be eliminated with a resultant decrease of within-treatment variability and increased precision.
- Basis of the Rule
  - Total variability in a population can be divided into two components: among blocks and within blocks. By assigning treatments within blocks, between block variability is eliminated.

Design, conduct, and analysis

- Rule of Thumb
  - The effects of two or more factors can be assessed simultaneously by means of factorial designs in which the treatment combinations are applied to the observation units.
  - Higher-order interactions occur rarely.
  - Analysis should follow design. Sampling strategy used to generate the data determines the appropriate analysis-tying the two together tightly.

Consulting

- Rule of Thumb
  - Structure a consulting session to have beginning, a middle, and an end. The session begins with problem context and definition, moves to resolution and solution, and concludes with a summary and allocation of responsibilities.
- Basis of the Rule
  - Consultation involves process and substance. It is important that the consultant be aware of the process and be able to implement it. Structuring the consulting session with a clear steps and product in view will pace guide the process.

Consulting

- Rule of Thumb
  - Pace the consultation by asking focused , helpful questions.
- Basis of the Rule
  - Asking questions show active listening, paces the conversation, allows the investigator to reflect, and may turn up surprising answers.

Consulting

- Rule of Thumb
  - Make valid distinctions, not distinctions valid.

Table 10.1 Basic Distinctions to Be Made in a Consulting Session

Item	Choice I	Choice II
Intent of study	Exploratory	Confirmatory
Endpoint	Measurable	Not well defined
Source of data	Random	Convenience
Sampling unit	Simple	Multiple
Randomization	Single	Multiple
Model	Fixed	Random or mixed
Type of study (1)	Intrinsic	Comparative
Type of study (2)	Cross-sectional	Longitudinal
Epidemiologic study	Cohort	Case-Control
Categories	Nominal	Some kind of ordering
Numerator/denominator	Rate	Proportion, ratio

Consulting

- Rule of Thumb
  - Know yourself, and know the investigator.
- Basis of the Rule
  - Investigators have a way of probing the limits of the consultant's knowledge. In fact, they frequently come because they were not able to apply the standard examples found in many textbooks.

Consulting

- Rule of Thumb
  - Tailor advice to the investigator's level of understanding. Avoid jargon, equations, and formulas.
- Basis of the Rule
  - Communication is important!

Consulting

- Rule of Thumb
  - When communicating with nonstatisticians, use natural units as much as possible.
- Basis of the Rule
  - It has been said that probability theory is hard-even for probabilities. This is doubly true for nonprobabilists. A “natural” unit is not easy to define explicitly, but it basically refers to a valid unit closest to the investigator’s experience and comfort zone.

Consulting

- Rule of Thumb
  - When the consulting session is done, be clear as to who does what, and when.
- Basis of the Rule
  - Consulting is like raising children. Everyone is happier if the rules and limits are clear.

Consulting

- Rule of Thumb
  - Any basic statistical package will do, any textbook will do.
- Basis of the Rule
  - Most standard statistical packages have been tested and are reliable.

Consulting

- Rule of Thumb
  - Ethics precedes, guides, and follows consultation.
- Basis of the Rule
  - Ethics shapes the context of a consultation. While many aspects of the context are understood or part of the ethos of the scientific community, there are basic values that may have to be enunciated before collegial consulting can take place.

Consulting

- Rule of Thumb
  - Be proactive. Being proactive means taking the lead in scientific discussions, coordinating separate efforts, and thinking broadly about the scientific enterprise.
- Basis of the Rule
  - Statistical consulting is best done at the start of research or production projects. By being proactive the statistician indicates commitment to such projects.

Consulting

- Rule of Thumb
  - Use the WEB for reference, resource, and education-for consultants and investigators. BUT USE IT WITH DISCERNMENT!
- Basis of the Rule
  - Somewhere on the WEB is a reliable, authoritative answer to most of the basic statistical consulting questions. The trick is to find it, evaluate it, and apply it.

Consulting: Listen to, and heed the advice of experts in the field

- If possible collaborate.
- Frequently review what is being done to check that the statistical analysts addresses the correct questions.
- Aim, if feasible, to see some raw data, to understand the measurement processes involved, have some appreciation of the general quality of the data.
- Enquire into aspects of the study design that might have bearing on the appropriate analysis.
- Begin with very simple methods.

Consulting: Listen to, and heed the advice of experts in the field

- If possible, end with simple methods.
- Since nice ideas for analysis often do not work the first time, be prepared to do some modification.
- Do not be frightened to make strong assumptions. When a preliminary answer has been obtained, then consider which of the assumptions may be crucial.
- Take considerable care over presentation and conclusion.
- If your work is to be acknowledged in a paper or a report, ask firmly to see what is written before it is submitted.

Consulting: Listen to, and heed the advice of experts in the field

- If you feel you should have been a co-author and have not been invited to be, pause for a few days.
- Occasionally, very rarely one hopes, be prepared to say that the data are incapable of throwing useful lights of the issues involved.
- Find a good balance between thinking things out for yourself and obtaining advice from statistical colleagues (finding time to help them in return).
- If more than 10% of what you do ends up by being directly useful, you are doing well.
- If the investigator begins by saying he has a trivial little problems which he is sure you will be able to sort out immediately, don't altogether BELIEVE him!

END OF RULES OF THUMB